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Front end conveyor having integrated transverse scrap conveyor

Abstract

A front end conveyor includes a frame having a first side and a second side, a lower deck and an upper deck. The lower deck includes a plurality of sheet supports, and upper portions of the sheet supports lie in a first plane. The upper deck includes a plurality of sheet guides, and the plurality of sheet supports and the plurality of sheet guides define therebetween a sheet transport path for moving sheets in a sheet transport direction toward the downstream end of the front end conveyor. A transverse scrap conveyor is mounted below the lower deck to carry scrap in a direction perpendicular to the sheet transport direction. The transverse scrap conveyor is secured to and supported by the frame such that the transverse scrap conveyor is movable with the frame as a unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of U.S. provisional patent application No. 63/183,889 filed May 4, 2021, the entire contents of which is hereby incorporated by reference.

TECHNICAL FIELD

(1) The present disclosure is directed to a front end conveyor for receiving sheets of material from a rotary die cut machine, and, more specifically, toward a front end conveyor having an integrated a transverse scrap conveyor.

BACKGROUND

(2) A layboy is conventionally used to receive sheets of material, often corrugated paperboard, from a rotary die cut (RDC) machine and to feed the sheets to a downstream conveyor such as a stacker. Layboys conventionally comprise a set of upper belts arranged above a set of lower belts to define therebetween a sheet transport pathway. Sheets are received at an upstream nip at the end of the

layboy that faces the rotary die cut machine and exit from a downstream nip that faces the input end of a downstream stacker or other conveyor.

(3) Instead of a layboy, a machine that may be referred to as a “front end conveyor,” a “front end system” or merely as a “front end” may sometimes be used. The front end conveyor is also intended to be positioned downstream of a rotary die cut machine and upstream of another conveyor such as a stacker. Unlike a layboy, a front end conveyor may have a sheet transport path that is defined by wheels in place of the upper and/or lower belts. The wheels are configured to make direct contact with the moving sheets. An example of a front end conveyor is disclosed in U.S. Pat. No. 10,071,873, assigned to the present applicant, which is hereby incorporated by reference.

(4) One benefit provided by front end conveyors is an improved ability to remove scrap material from sheets of material being transported through the front end conveyor relative to a conventional layboy. See, for example, the system disclosed in U.S. Pat. No. 9,027,737 which is also incorporated herein by reference. However, known front end conveyors are not always able to remove one hundred percent of the scrap attached to or associated with the sheets being transported.

(5) A transverse scrap conveyor is sometimes placed between a rotary die cut machine and a layboy or a front end conveyor to catch scrap material that drops from the rotary die cut machine or that falls near the upstream end of the front end conveyor. A conventional transverse scrap conveyor sits on the ground directly downstream from the rotary die cut machine. Its purpose is to transfer scrap to one side, often the drive side, of the layboy or front end conveyor for disposal. Because this conveyor is typically fixed in place, it can be a difficult to work around when operators need to separate the front end conveyor and the rotary die cut machine to change or service cutting dies of the rotary die cut machine.

(6) Sometimes transverse scrap conveyors are mounted in a pit in the ground immediately downstream of the rotary die cut machine. This may make the scrap conveyor easier to step over; however, the presence of a pit in a facility introduces another set of problems, namely, the pit must be marked and guarded to reduce the likelihood of persons falling into the pit. Also, the pit must be filled in and a new pit dug if a production line is moved. Pits are generally considered highly undesirable.

(7) It would therefore be desirable to provide a front end conveyor having improved scrap removal capabilities and that avoids or reduces the above-mentioned problems.

SUMMARY

(8) These and other problems are addressed by embodiments of the present invention, a first aspect of which comprises a front end conveyor comprising a frame having a first side and a second side, a lower deck and an upper deck. The lower deck includes a plurality of sheet supports, and upper portions of the sheet supports lie in a first plane. The upper deck includes a plurality of sheet guides, and the plurality of sheet supports and the plurality of sheet guides define therebetween a sheet transport path for moving sheets in a sheet transport direction toward the downstream end of the front end conveyor. A transverse scrap conveyor is mounted below the lower deck to carry scrap in a direction perpendicular to the sheet transport direction, and the transverse scrap conveyor is secured to and supported by the frame such that the transverse scrap conveyor is movable with the frame as a unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a left-front perspective view of a front end conveyor according to a first embodiment of the present disclosure.

- (2) FIG. 2 is a right-front perspective view of the front end conveyor of FIG. 1.
- (3) FIG. 3 is a left side elevational view of the front end conveyor of FIG. 1 with side panels removed to show the interior of the conveyor.
- (4) FIG. 4 is a front elevational view of the front end conveyor of FIG. 1.
- (5) FIG. 5 is a left-front perspective view of the front end conveyor of FIG. 1 with an upstream section and a downstream section of the upper conveyor deck raised.
- (6) FIG. 6 is a left side elevational view of the upstream end of the lower deck of the front end conveyor of FIG. 1 with an entry wheel in a raised position.
- (7) FIG. 7 is a right side elevational view of the upstream end of the lower deck of the front end conveyor of FIG. 1 with the entry wheel of FIG. 6 in the raised position.
- (8) FIG. 8 is a left side elevational view of the upstream end of the lower deck of the front end conveyor of FIG. 6 with an entry wheel in a lowered position.
- (9) FIG. 9 is a right side elevational view of the upstream end of the lower deck of the front end conveyor of FIG. 7 with the entry wheel of FIG. 7 in the raised position.
- (10) FIG. 10 is a top plan view of an actuating mechanism for shifting the entry wheel between the raised and lowered positions.
- (11) FIG. 11 is a sectional elevational view taken along line XI-XI in FIG. 10.
- (12) FIG. 12 is a detail view of region XII in FIG. 11.
- (13) FIG. 13 is left side elevational view of a front end conveyor according to a second embodiment of the present disclosure which includes a helical brush for scrap removal.
- (14) FIG. 14 is a perspective view of the helical brush of FIG. 13.
- (15) FIG. 15 is a left side elevational view of a front end conveyor according to a third embodiment of the present invention which includes a movable chad wall.
- (16) FIG. 16 is a perspective view of the movable chad wall of FIG. 15.
- (17) FIG. 17 is a left side elevational view of the front end conveyor of FIG. 15 with an air knife installed.
- (18) FIGS. 18 and 19 are right side perspective views of the front end conveyor of FIG. 1 partially exploded to show elements of the transverse scrap conveyor and its supports.
- (19) FIG. 20 shows a lift assembly for adjusting the height of the transverse scrap conveyor of FIGS. 18 and 19.

DETAILED DESCRIPTION

(20) Referring now to the drawings, wherein the showings are for purposes of illustrating presently preferred embodiments of the disclosure only and not for the purpose of limiting same, FIGS. 1-5 shows a front end conveyor 10 having an upstream end 12, a downstream end 14, a controller 15, which may include a microprocessor, a frame 16, and an electrical cabinet 17. The side of the front end conveyor 10 closest to the viewer in FIG. 1 may be referred to as the “first” or “left” side of the conveyor 10; the side of the conveyor 10 closest to the viewer in FIG. 2 may be referred to as the “second” or “right” side of the frame. Various features and configurations of this front end conveyor 10 are discussed below.

(21) Dual Clamshell Configuration

(22) The frame 16 supports a lower deck 18, an upstream upper deck 20 and a downstream upper deck 22. The lower deck 18 comprises a plurality of transverse shafts 24 each of which supports a plurality of disks 26 for rotation with the transverse shafts 24, and the upper portions of the annular outer surfaces of the disks 26 lie in a first plane and form a lower boundary of a sheet transport path 28 (FIG. 3) through the front end conveyor 10. The disks 26 may, for example, comprise aluminum wheels with polyurethane covers. The transverse shafts 24 and their disks 26 may sometimes be referred to as “sheet supports.” As shown in FIG. 2, the transverse shafts 24 each include an end pulley 25, and a serpentine belt 27 winds around the end pulleys 25 and a plurality of idler pulleys 29. A drive (not shown) is operatively connected to the belt 27 such that driving the belt 27 causes all the transverse shafts 24 and the attached disks 26 to rotate.

(23) The upstream upper deck **20** and the downstream upper deck **22** each comprise a plurality of belts **30**, and each belt **30** is supported by at least two pulleys **32**. The belts **30** and pulleys **32** may be referred to as “sheet guides.” The lower surfaces of the belts **30** lie in a second plane and define an upper boundary of the sheet transport path **28**.

(24) The spacing between the first plane and the second plane may be adjusted by, for example, raising or lowering the upper decks **20**, **22** relative to the lower deck **18**. Furthermore, while the second plane is generally located above the first plane during operation of the front end conveyor **10**, in some situations it may be useful to position the second plane at or below the first plane such that the bottoms of the belts **30** are located below the tops of the disks **26** to flex sheets of material in the sheet transport path **28** for improved control of certain types of materials. In this case, the sheet transport path **28** will still be the path followed by sheets moving through the front end conveyor **10** but the path will undulate in the transverse direction, passing under each of the belts **30** and over each of the disks **26**.

(25) The upstream upper deck **20** includes an upstream end **34** and a downstream end **36**, and the downstream upper deck **22** includes an upstream end **38** at the downstream end **36** of the upstream upper deck **20** and a downstream end **40**. An upstream pivot shaft **42** extends along the downstream end **36** of the upstream upper deck **20**, or, in the alternative, the upstream pivot shaft **42** may comprise two shorter individual pivot shafts at opposite sides of the upstream upper deck **20**. A downstream pivot shaft **44** extends along the upstream end **38** of the downstream upper deck **22**, or, in the alternative, the downstream pivot shaft **44** may also comprise two shorter individual pivot shafts at opposite sides of the downstream upper deck **22**. The ends of the upstream pivot shaft **42** are received in semicircular bearings **46** on opposite sides of the frame **16**, and the ends of the downstream pivot shaft **44** are received in semicircular bearings **48** on opposite sides of the frame **16** adjacent to the bearings **46**. This arrangement allows the upstream upper deck **20** to be pivoted about the axis of the upstream pivot shaft **42** to raise the upstream end **34** of the upstream upper deck **20** relative to the upstream end of the lower deck **18** and also allows the downstream end **40** of the downstream upper deck **22** to be pivoted around the axis of the downstream pivot shaft **44** to raise the downstream end **40** relative to the downstream end of the lower deck **18**.

(26) While the upper decks **20**, **22** can be raised and lowered manually, preferably electric actuators **49** are provided at the left and right sides of the upstream end **34** of the upstream upper deck **20** and at left and right sides of the downstream end **40** of the downstream upper deck **22** to raise and lower the upstream end **34** of the upstream upper deck **20** and the downstream end **40** of the downstream upper deck **22** and to secure the upper decks **20**, **22** in a raised or lowered position. Other mechanisms, such as gas springs, can alternately be used to hold the upper decks **20**, **22** in a raised position. Stops (not illustrated) may be provided to limit the downward movement of the upstream end **34** of the upstream upper deck **20** and the downstream end **40** of the downstream upper deck **22** to positions in which the bottoms of the belts **30** of the upstream upper deck **20** and the bottoms of the belts **30** of the downstream upper deck **22** lie in the same plane above the lower deck **18**.

(27) In operation, with reference to FIG. 3, sheets of material **54** exit the output **56** of a rotary die cut machine **58** and enter the upstream end of the sheet transport path **28** of the front end conveyor **10**. The transverse shafts **24** of the lower deck and/or the belts **30** of the upper deck are driven by one or more drives (not illustrated) to pull the sheets **54** into the front end conveyor **10** and convey the sheets **54** from the upstream end **12** to the downstream end **14** of the front end conveyor **10**. The sheets **54** exit the downstream end **14** of the front end conveyor **10** to a stacker **58** or other downstream conveyor.

(28) When it is desirable to access the interior of the front end conveyor **10**, e.g., to remove accumulated scrap or jammed sheets of material, or to perform maintenance or belt replacement, the actuators **49** are operated by the controller **15** to extend and thus raise the upstream end **34** of the upstream upper deck **20** relative to the upstream end of the lower deck **18** and/or to raise the

downstream end **40** of the downstream upper deck **22** relative to the downstream end of the lower deck **18** and to hold the upper decks **20, 22**. The upstream upper deck **20** and the downstream lower deck **22** can be raised and lowered at the same time or independently of one another. The actuators **49** also hold the upstream upper deck **20** and the downstream upper deck **22** in their raised positions while necessary maintenance is performed on elements in the interior of the front end conveyor **10**. The two pivotable upper decks **20, 22**, allow a worker to access all portions of the interior of the front end conveyor **10** from either the upstream end **12** or the downstream end **14** without having to reach over more than one half the length of the front end conveyor **10**. When work on the interior of the front end conveyor **10** is complete, the controller **15** operates the electric actuators **49** to retract and thus lower the upstream end **34** of the upstream upper deck **20** and the downstream end **40** of the downstream upper deck **22** until the bottoms of the belts **30** of the upstream upper deck **20** and the downstream upper deck **22** again lie in the second plane.

(29) Entry Drop-Down Wheel Shaft

(30) The front end conveyor **10** includes a feature that may be referred to as an entry drop down wheel shaft or a pivotable sheet support **50** which can be adjusted relative to the plane of the sheet transport path **28**. The ability to change the position of the pivotable sheet support **50** provides increased control over the entry of the front end conveyor **10**, either to increase the space for scrap to fall or to increase the width of the “jaws” of the front end conveyor **10** to help guide the sheets of material **54** into the front end conveyor **10** from the rotary die cut machine **58**.

(31) FIGS. **6** and **8** show the left side of the lower deck **18** with the pivotable sheet support **50** in a raised position; FIGS. **7** and **9** show the right side of the lower deck **18** with the pivotable sheet support **50** in a lowered position. The pivotable sheet support **50** comprises a plurality of frame members **51** each having a first opening **52** through which the upstream-most transverse shaft **24** of the primary sheet supports passes. An entry shaft **60** passes through second openings **53** in the frame member **51** and is rotationally supported in these second openings **53**. A plurality of entry disks **62** are mounted on the entry shaft **60** for rotation therewith. All of the transverse shafts **24** are driven by the serpentine belt **27** at the right side of the shafts **24** as discussed above and shown in FIG. **2**. A pulley **63** (FIG. **1**) at the left side of the upstream-most one of the transverse shafts **24** is connected to a pulley **64** at the left side of the entry shaft **60** by a belt (not shown) so that the rotation of the upstream-most one of the transverse shafts **24** rotates the entry shaft **60** and its associated entry disks **62**. The entry shaft **60** is thus operatively coupled to the upstream-most transverse shaft **24** and rotates when the upstream-most transverse shaft **24** rotates when the pivotable sheet support **50** is in the raised position of FIGS. **6** and **7** or in the lowered position of FIGS. **8** and **9** or in any intermediate position therebetween.

(32) An actuator **65**, which is preferably electric but which could also be hydraulic or pneumatic, is used to shift the entry shaft **60** between the raised position, in which tops of the entry disks **62** lie in the first plane and form an extension of the sheet transport path **28**, and the lowered position in which the entry disks **62** are located entirely below the first plane. The connection between the actuator **65** and the pivotable sheet support **50** is described below.

(33) Referring now to FIG. **10**, a plurality of rails **66** are mounted to the frame **16** such that they extend in the sheet transport direction. As best seen in FIG. **11** and the detail view of FIG. **12**, a slide **67** is slidably mounted on each of the rails **66**, and a rectangular plate **68** having a longest dimension in the sheet transport direction is connected to each of the slides **67**. A pivot arm **69** has a first end pivotally connected to the plate **68** and a second end pivotally connected to the frame member **51**.

(34) A rack **70** depends from each of the plates **68**, and a timing shaft **73** extends transversely to the sheet transport direction and is rotatably supported by the frame **16**. A plurality of timing gears **74** are mounted to the timing shaft **73** for rotation therewith. Each of the timing gears **74** engages one of the racks **70**. The timing shaft **73** is not powered but rather connects the slides **67** to each other and ensures that the slides **67** move synchronously along the rails **66**.

(35) As shown in FIG. 10, a piston 75 of the actuator 65 is connected to the plate 68 on the rail 66 at the left side of the front end conveyor 10 by a connector 76. In FIG. 10, the piston 75 is in its extended position, and the pivotable sheet support 50 is in its raised position. When the actuator 65 is retracted (under the control of the controller 15), it pulls the plate 68 and its associated slide 67 and rack 70 away from the upstream end 12 of the front end conveyor 10. The connection between the slides 67 provided by the racks 70 and the timing shaft 72 and timing gears 74 ensures that the linear force provided by the actuator 65 is applied evenly to the each of the slides 67 so that the slides 67 move smoothly and do not experience lateral forces. This allows a single actuator 65 to be used to actuate the pivotable sheet support 50. As the moving plates 68 pull the second ends of the pivot arms 68 in the downstream direction, the entry shaft 60 supported by the frame members 51 pivots about the axis of the upstream-most transverse shaft 24 to the lowered position.

(36) A plane tangent to the disks 26 of the upstream-most transverse shaft 24 and the entry disks 62 of the entry shaft 60 may be referred to as a third plane. When the pivotable sheet support 50 is in the raised position, the third plane is coextensive with the first plane. When the pivotably sheet support 50 is in the lowered position, the third plane intersects the first plane along a line transverse to the sheet transport direction.

(37) Helical Brush

(38) FIGS. 13-15 show a brush assembly 80 configured to remove scrap from the tops of sheets of material 54 traveling along the sheet transport path 28. The brush assembly 80 includes a transverse beam 82 that is supported by the frame 16, a drive 84 mounted to the transverse beam 82, a plurality of belt housings 86 that extend downwardly from the transverse beam 82 and a drive shaft 88 that is operatively connected to the drive 84 and which extends through and is rotationally supported in aligned openings 90 in upper portions of the belt housings 84.

(39) A brush shaft 92 is rotationally supported at the lower end of each of the belt housings 84 such that a portion of the brush shaft 92 projects from each side of each belt housing 84, and a helical brush segment 94 is mounted on each end of each of the brush shafts 92 for rotation therewith. The two brush segments 94 and their associated brush shafts 92 may be referred to as a “helical brush 96,” and FIG. 14 shows four helical brushes 96 each including two brush segments 94. A drive belt or drive chain 98 (referred to generically as a “drive belt”) is located inside each of the belt housings 84 and connects the drive shaft 88 to each of the brush shafts 92 such that rotation of the drive shaft 88 drives the drive belts 98 and thus rotates the brush shafts 92 and the helical brush segments 94.

(40) The helical brush segments 94 on a given brush shaft 92 are spaced apart from each other along the brush shaft 92, and the belt housings 84 support the brush shafts 92 at a location between the brush segments 94 on a given brush shaft 92. Also the brush shafts 92 of each helical brush 96 are spaced from one another. When the helical brushes 96 rotate, they apply forces against any scrap on the top surfaces of the sheets of material 54 in both longitudinal and transverse directions (relative to the longitudinal movement of the sheets of material 54 along the sheet transport path 28). This helps to both dislodge the scrap and to move the scrap toward the sides of the sheets of material 54 (transverse to the sheet transport direction) where the scrap can fall through the lower deck 18 to the ground or onto a longitudinal scrap conveyor 100.

(41) The speed, rotational direction and location of the helical brushes 96 relative to the sheet transport path 28 are adjustable. FIG. 13 shows a possible location for the helical brush assembly 80 above the sheet transport path 28. The brush assembly 80 can be located in a different position and/or more than one brush assembly 80 can be present. For example, a second helical brush assembly 80 could be located beneath the sheet transport path to brush against the lower sides of the sheets of material 54 in the sheet transport path 28.

(42) Moveable Chad Wall

(43) Small pieces of paperboard cut from a sheet of paperboard by the rotary die cut machine 58 may not be completely detached from the sheet of material 54 when it enters the front end

conveyor **10**. For example, small openings that are cut in the sheet of material **54** or the material removed from between flap portions of the sheet of material **54** may remain connected to the sheet by small connections. The punched pieces of material are referred to as “chads,” and chads that are not completely detached from the paperboard may be referred to as ‘hanging chads.’ A hanging chad **99** is illustrated in FIG. 3.

(44) It is important to remove both loose chads (not shown) and hanging chads **99** before the sheets of material **54** reach a final stack. The brush assembly **80** discussed above and the blower discussed below can remove loose chads from the upper surfaces of the sheets of material **54** traveling along the sheet transport path **18**. However, hanging chads **99**, especially when they hang downward from the sheets of material **54**, are more difficult to remove with a blower or a brush assembly **80**.

(45) In order to dislodge these hanging chads **99**, the disclosed front end conveyor **10** includes a chad wall **110** which is illustrated in FIGS. 15-17. The chad wall **110** comprises a rectangular sheet of material, such as sheet metal or rigid plastic, that has a width approximately equal to the width of the sheet transport path **28**. The chad wall **110** includes a top edge **112**, a front surface **114** and a plurality of fingers **116** that project from the front surface **114** from locations along the top edge **112**. The front surface **114** is the surface that faces the upstream end **12** of the front end conveyor **10**, and the fingers **116** thus extend toward the sheets of material **54** traveling along the sheet transport path **28**. The fingers **116** have planar top surfaces **118**, and the top surfaces **118** are angled slightly downward in the upstream direction.

(46) The chad wall **110** also includes a plurality of vertical slits **120** via which the chad wall **110** is adjustably mounted to a portion of the frame **16** of the front end conveyor **10** by a plurality of bolts or similar fasteners **122**. Loosening the fasteners **122** allows the chad wall **110** to be slid up and down to place the top edge **112** a desired distance below the sheet transport path **28** at which time the fasteners **122** can be tightened to hold the chad wall **110** in a desired position.

(47) The location of the top edge **112** of the chad wall **110** relative to the sheet transport path **28** can be seen in FIG. 15. During operation of the front end conveyor **10**, the sheets of material **54** travel along the sheet transfer path **28**. Planar portions of the sheets **54** that remain supported at or above the upper portions of the disks **26** of the lower deck **18** will not contact any portion of the chad wall **110**. If the front portions of the sheets of material **54** droop a small amount, for example, if flaps are present at the front edges of the sheets of material **54**, these front portions may drop below the level of the top edge **112** of the chad wall **110**. These drooping front portions will contact and slide up the top surfaces **118** of the fingers **116** so that the sheets of material **54** pass over the chad wall **110** without damaging the sheets of material **54** or adversely affecting the movement of the sheets of material **54** through the front end conveyor **10**.

(48) Any hanging chads **99** extending below the level of the top edge **112** of the chad wall **110**, however, will either impact the front surface **114** of the chad wall or against one of the fingers **116**. Hanging chads **99** impacting the front surface **114** of the chad wall **110** will likely be torn from the moving sheets **54** due to the speed of the moving sheets **54**. Hanging chads **99** that impact off-center on one of the fingers **116** may be torn and/or twisted and dislodged from the sheets of material **54**. Finally, hanging chads **99** that happen to be aligned with one of the fingers **116** may impact squarely against that finger **116**. However, the impact against the leading edge of the finger **116** in combination with the rapid folding of the hanging chad **99** upwardly against the underside of the sheet of material **54** as the hanging chad **99** travels along the top surface **118** of the finger **116** is also likely to dislodge the hanging chad **99**. The chad wall **110** therefore significantly reduces the number of hanging chads **99** on the sheets of material **54** as the sheets of material **54** move through the front end conveyor **10**.

(49) Blower/Air Knife System

(50) FIG. 17 shows a blower in the form of an air knife **130** mounted above the sheet transport path **54**. The air knife **130** extends across substantially the entire width of the sheet transport path **54** and is directed at the sheet transport path **54** to dislodge scrap material present on the sheets of material

54. The air knife **130** may have a single outlet that extends across substantially the entire width of the sheet transport path or alternately may include a plurality of separate jets, the direction of each of which may be independently adjustable. When the blower is sufficiently powerful or when an air knife **130** is used as the blower, even chads that remain partially attached to the upper surfaces of the sheets of material **54** may be severed or torn from the sheets of material **54** and blown off the sheets of material **54**.

(51) FIG. **17** shows the air knife **130** located about half way between the upstream and downstream ends **12**, **14** of the front end conveyor **10** below the junction of the upstream upper deck **20** and the downstream upper deck **22**. The upstream upper deck **20** includes a first set of belts **30** and the downstream upper deck **22** includes a second set of belts **30** spaced apart from the belts **30** of the upstream upper deck **20**. Therefore no belts **30** contact the sheets of material **54** in the location where the air knife **130** is mounted. It is preferable to place the air knife **130** in this location (or at least to direct the air output of the air knife **130** toward this location) where no belts **30** are present to hold scrap against the upper surfaces of sheets of material **54**. This location increases the likelihood that scrap will be detached or at least loosened from the sheets of material **54** if not completely removed by the air knife **130**. The loosened scrap material may then be more easily removed by the helical brush assembly **80** discussed above. If desired, a second blower or air brush **130** can be placed beneath the sheet transport path and directed against the lower surfaces of the sheets of material **54** to help dislodge hanging chads **99**.

(52) The air knife **130** is preferably rotatable about an axis transverse to the sheet transport direction so that the output of the air knife can be directed at different locations along the sheet transport path. The air knife is also preferably mounted so that it can be raised and lowered relative to the sheet transport path.

(53) Integrated Transverse Scrap Conveyor

(54) The present front end conveyor **10** also includes a transverse scrap conveyor **140** that can be attached to the frame **16** of the front end conveyor **10** so that the transverse scrap conveyor **140** is movable with the front end conveyor **10**. By integrating the transverse scrap conveyor **140** with the front end conveyor **10** in this manner, the transverse scrap conveyor **140** is movable with the front end conveyor **10** when an operator needs to access the rotary die cut machine **58**, and the operator therefore does not need to reach over or move a stand-alone transverse scrap conveyor as has been done in the past. Beneficially, the transverse scrap conveyor **140** can also be detached from the front end conveyor and used as a stand-alone unit if desired.

(55) The transverse scrap conveyor **140** and its associated supports can be seen for example, in FIGS. **1** and **5**. However, the following description is directed to FIGS. **18** and **19** which show the transverse scrap conveyor **140** partially separated from the front end conveyor **10** to make various components thereof easier to see.

(56) The transverse scrap conveyor **140** includes a conveyor frame **142** that supports a driven end roller **144** and a non-driven end roller **146**. A belt **148** extends around the driven end roller **144** and the non-driven end roller **146**, and a drive **150** is operatively connected to the driven end roller **144** to rotate the driven end roller **144** to move the belt **148** in a forward or reverse direction around the conveyor frame **142**.

(57) The transverse scrap conveyor **140** is mountable in a cradle **152** that includes an upstanding front wall **154**, a rear wall **156** and channel members **158** that connect the front wall **154** to the rear wall **156**. The distance from the front wall **154** of the cradle **152** to the rear wall **156** of the cradle **152** is approximately the same as the width of the transverse scrap conveyor **140**, and the transverse scrap conveyor **140** is mounted in the cradle **152** and connected to the conveyor frame **142** by bolts or other fasteners that extend through the front wall **154** of the cradle **152** into one side of the conveyor frame **142** and through the rear wall **156** of the cradle **152** into another side of the conveyor frame **142**. An angled scrap guide **159** is located at the top edge of the front wall **154**.

(58) The cradle **152** is mountable to a support frame **160** that includes a body member **162** that

extends in the transverse direction of the front end conveyor **10** and two arm supports **164** that project from the body member **162** in the upstream direction. The bottoms of the arm supports **164** are located above the support surface on which the front end conveyor **10** rests when the body member **162** is connected to the frame **16**. The arm supports **164** have a profile complementary to bottom openings of the channel members **158** which allows the channel members **158** to receive the arm supports **164** to securely support the cradle **152** on the support frame **160**. The cradle **152** is also bolted to the support frame **160**.

(59) Mounting the transverse scrap conveyor **140** to the front end conveyor **10** in this manner integrates the transverse scrap conveyor **140** and the front end conveyor **10** so that the transverse scrap conveyor **140** moves as a component of the front end conveyor **10** and thus moves out of the way with the front end conveyor **10** when the front end conveyor **10** is rolled away from the rotary die cut machine **58** when it is necessary to access the side of the rotary die cut machine that faces the front end conveyor **10**.

(60) FIG. **20** is an exploded view of the cradle **152** and an optional lift system **166** for the cradle **152** that allows the cradle **152** to rest on the support surface that supports the front end conveyor **10** when it is not desired to connect the transverse scrap conveyor **140** to the support frame **160** of the front end conveyor **10**. The lift system **166** includes four adjustable supports **168** each mounted at a corner of the cradle **152**, and each support **168** includes a cylinder **170** and a foot **172** extending therefrom. The feet **172** can be manually screwed into and out of the cylinders **170** to adjust the height of the cradle **152** on the support surface. In addition, the feet **172** can be extended and retracted from the cylinders **170** by rotating a mechanism in an opening **176** in the cylinder **170**. The lift system **166** includes a first rod **178** connected between the openings **176** of a first pair of the supports **168** and a second rod **180** connected between the openings **176** of a second pair of the supports **168**. A drive **182** is provided for rotating a drive shaft **184**, and the drive shaft **184** is connected to the first rod **178** and the second rod **180** by two drive belts **186**. Operating the drive **182** in a first direction rotates the first rod **178** and the second rod **180** in a first direction to cause the feet **172** to move out of the cylinders **170**; operating the drive **182** in the opposite direction rotates the first rod **178** and the second rod **180** in a second direction to draw the feet **172** back into the cylinders **170**.

(61) When the feet **172** are fully retracted, the cradle **152** can be mounted to the support frame **160** and fully supported by the support frame **160**. The feet **172** can also be extended while the cradle **152** is mounted on the support frame **160** to provide additional stability to the transverse scrap conveyor **140** and retracted when it is desired to move the transverse scrap conveyor **140** as a unit with the front end conveyor **10**. Or, the supports **168** can provide independent support for the cradle **152** when it is not desired to connect the transverse scrap conveyor **140** to the front end conveyor **10**.

(62) In operation, scrap that falls through the bottom deck **18** lands on the longitudinal scrap conveyor **100** and is carried toward the upstream end **12** of the front end conveyor **10** and dropped onto the transverse scrap conveyor **140**. Scrap that falls from the rotary die cut machine **58** also drops onto the transverse scrap conveyor **140**, possibly guided by the angled scrap guide **159**. The transverse scrap conveyor **140** is driven to move the scrap to one side of the front end conveyor **10** from where it can be removed and discarded. When it is necessary to access the rotary die cut machine **58**, the front end conveyor **10** can be rolled away from the rotary die cut machine **58** and the attached transverse scrap conveyor **140** will move with the front end conveyor **10**.

(63) The present invention has been described herein in terms of several preferred embodiments. Modifications and additions to these embodiments will become apparent to persons of ordinary skill in the art upon a reading of the foregoing description. It is intended that all such modifications and additional form a part of the present invention to the extent they fall within the scope of the several claims appended hereto.

Claims

1. A front end conveyor having an upstream end and a downstream end, the front end conveyor comprising: a frame comprising a plurality of interconnected frame members, at least one first frame member of the plurality of interconnected frame members comprising a bottom frame member configured to rest on a support surface; a lower deck supported by the frame and including a plurality of sheet supports, upper portions of the sheet supports lying in a first plane; an upper deck supported by the frame and including a plurality of sheet guides, the plurality of sheet supports and the plurality of sheet guides defining therebetween a sheet transport path for moving sheets in a sheet transport direction toward the downstream end of the front end conveyor; and a transverse scrap conveyor below the lower deck configured to carry scrap in a direction perpendicular to the sheet transport direction, the transverse scrap conveyor being mounted on at least one second frame member of the plurality of frame members, wherein the at least one second frame member is connected to the at least one first frame member such that the at least one second frame member is held above the support surface when the bottom frame member rests on the support surface.
2. The front end conveyor according to claim 1, wherein the transverse scrap conveyor is located beneath an upstream end of the lower deck.
3. The front end conveyor according to claim 1, including a longitudinal scrap conveyor beneath the lower deck, the longitudinal scrap conveyor having a first end at the upstream end of the front end conveyor and a second end at the downstream end of the front end conveyor, the longitudinal scrap conveyor being configured to carry scrap that falls from the sheet transport path in the sheet transport direction or in a direction opposite the sheet transport direction, wherein the transverse scrap conveyor is located beneath the first end or the second end of the longitudinal scrap conveyor and configured to catch scrap that falls from the longitudinal scrap conveyor.
4. The front end conveyor according to claim 3, wherein the longitudinal scrap conveyor is configured to carry the scrap in the direction opposite the sheet transport direction, and wherein the transverse scrap conveyor is located beneath the first end of the longitudinal scrap conveyor.
5. The front end conveyor according to claim 4, including a drive configured to drive the transverse scrap conveyor, the drive being supported by the frame.
6. The front end conveyor according to claim 4, wherein the plurality of lower sheet supports comprise a plurality of shafts extending perpendicular to the sheet transport direction, and wherein a plurality of disks are attached to each of the plurality of shafts.
7. The front end conveyor according to claim 6, wherein the plurality of disks are configured to rotate with each of the plurality of shafts.
8. The front end conveyor according to claim 7, wherein the disks of a first shaft of the plurality of shafts are offset from the disks of a second shaft of the plurality of shafts in a direction transverse to the sheet transport direction.
9. The front end conveyor according to claim 8, wherein each of the disks has an annular outer surface, and wherein an upper portion of the outer surface of each of the disks is configured to contact the sheets of material in the sheet transport path.
10. The front end conveyor according to claim 9, wherein the upper sheet guides comprise a plurality of belts, and wherein each of the plurality of belts is supported by at least two pulleys.
11. A system comprising: a rotary die cut machine having an output; a stacking conveyor having an input; and the front end conveyor according to claim 1 arranged between the output of the rotary die cut machine and the input of the stacking conveyor and configured to receive the sheets of material from the rotary die cut machine at the upstream end of the front end conveyor and to output the sheets of material from the downstream end of the front end conveyor toward the input of the stacking conveyor.

12. The system according to claim 11, wherein the transverse scrap conveyor is positioned to catch scrap emitted from the output of the rotary die cut machine.
13. The front end conveyor according to claim 1, wherein the transverse scrap conveyor is supported by the frame such that the transverse scrap conveyor is movable with the frame as a unit.
14. A front end conveyor having an upstream end and a downstream end, the front end conveyor comprising: a frame comprising a plurality of interconnected frame members; a lower deck supported by the frame and including a plurality of sheet supports, upper portions of the sheet supports lying in a first plane; an upper deck supported by the frame and including a plurality of sheet guides, the plurality of sheet supports and the plurality of sheet guides defining therebetween a sheet transport path for moving sheets in a sheet transport direction toward the downstream end of the front end conveyor; and a transverse scrap conveyor below the lower deck configured to carry scrap in a direction perpendicular to the sheet transport direction, wherein a first frame member of the plurality of frame members is located below the lower conveyor deck and extends in the direction perpendicular to the sheet transport direction, wherein a second frame member of the plurality of frame members extends from the first frame member in an upstream direction, wherein a third frame member of the plurality of frame members extends from the first frame member in the upstream direction, wherein a cradle is supported by the second frame member and the third frame member, and wherein the transverse scrap conveyor is mounted in the cradle.
15. The front end conveyor according to claim 14, wherein the cradle includes at least one adjustable support extendable and retractable relative to the transverse scrap conveyor.
16. The front end conveyor according to claim 14, wherein the plurality of sheet supports comprise a plurality of shafts extending perpendicular to the sheet transport direction, and wherein a plurality of disks are attached to each of the plurality of shafts for rotation with each of the plurality of shafts.
17. The front end conveyor according to claim 14, wherein the cradle is mounted on the second frame member and the third frame member.
18. A front end conveyor having an upstream end and a downstream end, the front end conveyor comprising: a frame comprising a plurality of interconnected frame members including a first frame member and at least one cantilevered second frame member, the at least one second frame member extending from the first frame member in the upstream direction; a lower deck supported by the frame and including a plurality of sheet supports, upper portions of the sheet supports lying in a first plane; an upper deck supported by the frame and including a plurality of sheet guides, the plurality of sheet supports and the plurality of sheet guides defining therebetween a sheet transport path for moving sheets in a sheet transport direction toward the downstream end of the front end conveyor; and a transverse scrap conveyor at least partially supported by the at least one second frame member and configured to carry scrap in a direction perpendicular to the sheet transport direction.
19. The front end conveyor according to claim 18, wherein the a cradle is supported by the at least one second frame member, and wherein the transverse scrap conveyor is mounted in and supported by the cradle.
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